

1. Introduction to Data Integrity

- Data integrity refers to maintaining accuracy and consistency of data.
- It ensures data is reliable throughout its lifecycle.
- Example: A bank system must always store correct account balances.

2. Importance of Data Integrity

- Prevents data corruption and duplication.
- Ensures reliable decision-making.
- Improves database performance.
- Example: Wrong medical data can lead to incorrect treatment.

3. Types of Data Integrity

- Four main types:
 - 1. Entity Integrity
 - 2. Referential Integrity
 - 3. Domain Integrity
 - 4. User-defined Integrity
- Each plays a different role in maintaining correctness.

4. Entity Integrity Explained

- Primary key uniquely identifies each record.
- Cannot be NULL or duplicate.
- Example:
- Student(ID, Name)
- ID must be unique.

5. Entity Integrity Example

- Table:
- ID | Name
- 1 | Ali
- 2 | Sara
- Invalid:
- ID = NULL or duplicate ID.
- Ensures every row is uniquely identifiable.

6. Referential Integrity Explained

- Maintains relationships between tables.
- Foreign key must match primary key.
- Prevents invalid references.

7. Referential Integrity Example

- Customers(ID)
- Orders(CustomerID)
- CustomerID in Orders must exist in Customers.
- Invalid: Order with CustomerID that does not exist.

8. Domain Integrity Explained

- Ensures values fall within valid range.
- Defined using data types and constraints.

9. Domain Integrity Example

- Age must be between 1 and 120.
- Marks must be between 0 and 100.
- Invalid: Age = -5 or Marks = 150.

10. User-defined Integrity

- Custom constraints defined by user.
- Depends on business rules.

11. User-defined Integrity Example

- Salary must be > 5000 .
- Student must enroll in at least one course.
- These rules are application-specific.

12. Introduction to Database Security

- Database security protects data from unauthorized access.
- Ensures confidentiality, integrity, and availability.

13. Importance of Security

- Protects sensitive information.
- Prevents cyber attacks.
- Maintains user trust.
- Example: Protecting user passwords.

14. Types of Threats

- Unauthorized access
- Data leakage
- SQL Injection
- Malware
- These threats can damage or steal data.

15. SQL Injection Example

- Input: ' OR '1'='1
- Can bypass login systems.
- Attackers gain unauthorized access.

16. Authentication

- Verifies identity of users.
- Methods: Passwords, biometrics.
- Example: Login system.

17. Authorization

- Defines what users can do.
- Example:
- Admin: full access
- User: read-only access.

18. Encryption

- Converts data into unreadable form.
- Only authorized users can decrypt it.
- Example: Encrypted passwords.

19. Backup

- Creates copies of database.
- Used to prevent data loss.

20. Recovery

- Restores data after failure.
- Example: Recovering after system crash.

21. Integrity vs Security

- Integrity ensures correctness.
- Security ensures protection.
- Both are essential.

22. Real-life Example

- Bank system:
- Integrity: correct balance
- Security: only owner can access account.

23. Summary

- Data integrity ensures valid data.
- Security protects data from threats.
- Both are critical in DBMS.

24. Practice Question

- What happens if referential integrity is violated?
- Give one real-life example.

25. End Slide

- Thank You!
- Prepare well for exams.